

How to use these token cutouts

These pages contain the text *Green Eggs and Ham* by Dr Seuss. The **red text** above each token shows its prefix—the 2 words before it in the original text.

Training: build the cutouts model

1. **cut out the tokens** from the sheets above along the dotted lines (carefully—keep each prefix-token pair together)
2. **spread them out** on a table, face up, no overlap if possible
3. that's your trained language model—every prefix-token pair from your training text is now a physical cutout on the table

Optional extension: once you've got the hang of it, group the cutouts into piles by their **red prefix label**. Each pile is a “bucket” in computer-science speak—the same trick a hash table uses to find values quickly.

Generation: sample new text

1. **choose a starting prefix**—any word that appears as a **red label** on at least one cutout
2. **write it down**, then make it your “current word”
3. **find candidates**—look across the spread for cutouts whose red prefix label matches your current word
4. **pick one** visually (your eye will land on cutouts more often when there are more of them—that's weighted sampling for free)
5. **write down** the picked cutout's token, then **put it back** in the spread
6. the token you just wrote is your new current word—go back to step 3

Tip: if a prefix has more cutouts in the spread, words following it are more common in the source text, and your eye will pick them more often.

Model statistics

- **Total tokens:** 940 (940 kept)
- **Entropy:** 0.95 bits/token — how unpredictable each pick is
- **Perplexity:** 1.9 — effective number of choices per pick

I	am	Sam	□	I	am	Sam	□
1	2	3	4	5	6	7	8
Sam	□	□	Sam	Sam	I	I	am
9	10	11	12	13	14	15	
Sam	I	am	□	that	Sam	I	
16	17	18	19	20	21	22	
am	that	Sam	I	am	I	do	
23	24	25	26	27	28		
I	do	do	not	like	that	Sam	I
29	30	31	32	33			
would	you	like	green	eggs			
34	35	36	37	38	39		
green	eggs	and	ham	I	do	not	like
40	41	42	43	44	45	46	
not	like	them	□	Sam	I	am	□
47	48	49	50	51	52		
do	not	like	green	eggs	and		
53	54	55	56	57	58		
eggs	and	ham	□	would	you	like	them
59	60	61	62	63	64		
like	them	here	or	there	I	would	not
65	66	67	68	69	70		
would	not	like	them	here	or	there	□
71	72	73	74	75			
there	□	□	I	would	not	like	them

like them

anywhere

76

them anywhere



77

anywhere



I

78

I

do

79

I do

not

80

do not

like

81

not like

green

82

like green

eggs

83

green eggs

and

84

eggs and

ham

85

and ham



86

ham



I

87

I

do

88

I do

not

89

do not

like

90

not like

them

91

like them



92

them



Sam

93

Sam



I

94

Sam I

am

95

I am



96

am



would

97

would



you

98

would you

like

99

you like

them

100

like them

in

101

them in

a

102

in a

house

103

a house

would

104

house would

you

105

would you

like

106

you like

them

107

like them

with

108

them with

a

109

with a

mouse

110

a mouse

I

111

mouse I

do

112

I do

not

113

do not

like

114

not like

them

115

like them

in

116

them in

a

117

in a

house

118

a house



119

house



I

120

I

do

121

I do

not

122

do not

like

123

not like

them

124

like them

with

125

them with

a

126

with a

mouse

127

a mouse



128

mouse



I

129

I

do

130

I do

not

131

do not

like

132

not like

them

133

like them

here

134

them here

or

135

here or

there

136

or there



137

there



I

138

I



do

139

I do

not

140

do not

like

141

not like

them

142

like them

anywhere

143

them anywhere



144

anywhere



I

145

I



do

146

I do not 147	do not like 148	not like green 149	like green eggs 150	green eggs and 151	eggs and ham 152
and ham 153	ham I 154	I do do 155	do not not 156	not like like 157	like them them 158
them Sam 160	Sam I 161	Sam I am 162	I am am 163	am would would 164	would you you eat 165
you eat them 167	eat them in 168	them in a 169	in a box 170	a box would 171	box would you 172
would you eat 173	you eat them 174	eat them with 175	them with a 176	with a fox 177	a fox not 178
fox not in 179	not in a 180	in a box 181	a box box 182	box not not 183	not with with 184
with a fox 186	a fox fox 187	fox not not 188	not in in 189	in a a house 190	a house house 191
house not not 193	not with with 194	not with a 195	with a mouse 196	a mouse mouse 197	mouse I I 198
I would would 199	I would not 200	would not eat 201	not eat them 202	eat them here 203	them here or 204
here or there 205	or there there 206	there I I 207	I would would 208	I would not 209	would not eat 210
not eat them 211	eat them anywhere 212	them anywhere anywhere 213	anywhere I I 214	I would would 215	would not eat 216
I would would 217	I would not 218	would not eat 219	not eat green 220	eat green eggs 221	

green eggs and ham I do not like
220 221 222 223 224 225 226

not like them like them them Sam I am
227 228 229 230 231 232

am would you could you in a
233 234 235 236 237 238

in a car eat them eat them here
239 240 241 242 243 244

them here they are I would not
245 246 247 248 249 250

would not could not in a car
251 252 253 254 255 256 257

a car car you you may like them you
258 259 260 261 262 263 264

you will see you may like
265 266 267 268 269 270

may like them in a tree I would
271 272 273 274 275 276

I would not could not in a tree
277 278 279 280 281 282 283

a tree tree not in a car you let me
284 285 286 287 288 289 290 291

let me be I do not like them
292 293 294 295 296 297 298

like them in 299	them in a 300	in a box 301	a box box  302	box  I 303	 I do 304	I do not 305	do not like 306
not like them 307	like them with 308	them with a 309	with a fox 310	a fox  311	fox  I 312	 I do 313	
I do not 314	do not like 315	not like them 316	like them in 317	them in a 318	in a house 319		
a house  320	house  I 321	 I do 322	I do not 323	do not like 324	not like them 325	like them with 326	
them with a 327	with a mouse 328	a mouse  329	mouse  I 330	 I do 331	I do not 332		
do not like 333	not like them 334	like them here 335	them here or 336	here or there 337	 338		
there  I 339	 I do 340	I do not 341	do not like 342	not like them 343	like them anywhere 344		
them anywhere  345	anywhere  I 346	 I do 347	I do not 348	do not like 349	not like green 350		
like green eggs 351	green eggs and 352	eggs and ham 353	and ham  354	ham  I 355	 I do 356	I do not 357	
do not like 358	not like them 359	like them  360	them  Sam 361	 Sam I 362	Sam I am 363	I am  364	
am  a 365	 a train 366	a train a 367	train a train 368	a train a 369	train a train 370	a train a 371	
train a train 372	a train could 373	train could you 374	 375	could you you  would 376			

would you 377	would you on 378	you on a 379	on a train 380	a train not 381	train not on 382	not on a 383
on a train 384	a train not 385	train not in 386	not in a 387	in a tree 388	a tree not 389	tree not in 390
not in a 391	in a car 392	a car Sam 393	car Sam let 394	Sam let me 395	let me be 396	me be I 397
be I would 398	I would not 399	would not would not 400	not would could 401	could not 402	could not could not 403	
not in in 404	in a a 405	in a box 406	a box box 407	box I I 408	I would would 409	I would not 410
would not would not 411	not could could 412	could not 413	could not could not 414	not with with 415	with a 416	
with a fox 417	a fox fox 418	fox I I 419	I will will 420	I will not 421	will not eat 422	not eat them 423
eat them in 424	them in a 425	in a house 426	a house house 427	house I I 428	I will will 429	I will not 430
will not eat 431	not eat them 432	eat them with 433	them with a 434	with a mouse 435	a mouse mouse 436	
mouse I I 437	I will will 438	I will not 439	will not eat 440	not eat them 441	eat them here 442	
them here or 443	here or there 444	or there there 445	there I I 446	I will will 447	I will not 448	will not eat 449
not eat them 450	eat them anywhere 451	them anywhere anywhere 452	anywhere I I 453	I do do 454		

I do not 455	do not eat 456	not eat green 457	eat green eggs 458	green eggs and 459	eggs and ham 460
and ham 461	ham I 462	I do do 463	I do not 464	do not like 465	not like them 466
like them 467	them Sam 468	Sam I I 469	Sam I am 470	I am say 471	am say in 472
say in the 473	in the dark 474	the dark here 475	dark here in 476	here in the 477	in the dark 478
the dark would 479	dark would you 480	would you 481	you could 482	could you 483	you in 484
in the dark 485	the dark I 486	the dark I 487	dark I would 488	I would not 489	would not 490
not could not 491	could not in 492	could not in 493	not in the 494	in the dark 495	the dark 496
dark would you 497	would you 498	would you could 499	you could you 500	could you 499	could you 501
you in the 502	in the rain 503	the rain I 504	rain I would 505	I would not 506	would not 507
would not 508	not could not 509	could not in 510	not in the 511	in the rain 512	the rain not 513
rain not in 514	not in the 515	in the dark 516	the dark not 517	dark not 518	not in a 519
on a train 520	a train not 521	train not in 522	not in a 523	not in a 524	in a car 525

a car □ 533	car □ not 534	□ not not in 535	not in in a 536	in a a tree 537	a tree tree □ 538	tree □ I 539	□ I do 540
I do not 541	do not like 542	not like them 543	like them □ 544	them □ Sam 545	□ Sam □ 546		
Sam □ you 547	□ you see 548	you see □ 549	see □ not 550	□ not in 551	not in a 552	in a house 553	
a house □ 554	house □ not 555	□ not in 556	not in a 557	in a box 558	a box □ 559	box □ not 560	
□ not with 561	not with a 562	with a mouse 563	□ □ 564	a mouse mouse □ not 565	mouse □ not 566	□ not with 567	
not with a 567	with a fox 568	a fox □ 569	fox □ I 570	□ I will 571	I will not 572	will not eat 573	
not eat them 574	eat them here 575	them here or 576	here or there 577	or there □ 578	there □ I 579		
□ I do 580	I do not 581	do not like 582	not like them 583	like them anywhere 584			
them anywhere you 585	anywhere you do 586	you do not 587	do not like 588	not like green 589			
like green eggs 590	green eggs and 591	eggs and ham 592	and ham I 593	ham I do 594	I do not 595	do not like 596	
not like them 597	like them □ 598	them □ Sam 599	□ Sam I 600	Sam I am 601	I am □ 602	am □ could 603	
□ could you 604	could you □ 605	you □ would 606	□ would you 607	would you □ 608	you □ with 609		

with a	with a goat	a goat I	goat I would	I would not	would not □		
not could	could not	could not with	not with a	with a goat	a goat would		
goat would you	would you □	you could	could you	could you □	you on		
on a	on a boat	a boat I	boat I could	I could not	could not □		
not would	could not	would not □	not on	on a	a boat □		
boat I	I will	I will not	will not □	not will	will not □		
not with	with a	with a goat	a goat I	goat I	I will	I will not	
will not eat	not eat them	eat them in	them in the	in the rain	the rain □	rain not	
not in	not in the	in the dark	the dark not	dark not in	not in a	in a tree	
a tree not	tree not in	not in a	in a car	a car you	car you let	you let me	let me be
me be I	be I do	I do not	do not like	not like them	like them in	them in a	
in a box	a box □	box I	I do	I do not	do not like	not like them	

like them with 691	them with a 692	with a fox 693	a fox fox 694	fox I 695	I will 696	I will not 697
will not eat 698	not eat them 699	eat them in 700	them in a 701	in a house 702	a house 	
house I 704	I do 705	I do not 706	do not like 707	not like them 708	like them with 709	
them with a 710	with a mouse 711	a mouse 	mouse I 713	I do 714	I do not 715	
do not like 716	not like them 717	like them here 718	them here or 719	here or there 720	or there 	
there I 722	I do 723	I do not 724	do not like 725	not like them 726	like them anywhere 727	
them anywhere 	anywhere I 728	I do 729	do not like 730	I do not 731	do not like 732	green 733
like green eggs 734	green eggs and 735	eggs and ham 736	and ham I 737	ham I do 738	I do not 739	do not like 740
not like them 741	like them 	them Sam 743	Sam I 744	Sam I am 745	I am 	am you 747
I you 748	you do not 749	do not like 750	not like them 751	like them 	them so 753	I so you 754
so you say 755	you say 	say try 757	I try 758	try them try 759	them try them 760	
try them and 761	them and you 762	and you may 763	you may 	may try 765	I try 766	

try them and 767	them and you 768	and you may 769	you may I 770	may I say 771	I say I 772	I say I 773
say Sam 774	Sam If 775	Sam If you 776	If you let 777	you let me 778	let me be 779	me be be I 780
I will 782	I will try 783	will try them 784	try them you 785	them you will 786	you will see 787	you will see 788
will see I 789	see I I 790	I I I 791	I I I 792	I and and 793	and he tries 794	and he tries 795
he tries them 796	tries them I 797	them I I 798	I I I 799	I say I 800	say I like 801	say I like 802
I like green 803	like green eggs 804	green eggs and 805	eggs and ham 806	and ham I 807	ham I do 808	I do I 809
do I like 810	I like them 811	like them I 812	them I Sam 813	I Sam I 814	Sam I am 815	Sam I am 815
I am and 816	am and I 817	and I would 818	I would eat 819	would eat them 820	eat them in 821	eat them in 821
them in a 822	in a boat 823	a boat I 824	boat I and 825	I and I 826	and I would 827	I would eat 828
would eat them 829	eat them with 830	them with a 831	with a goat 832	a goat I 833	goat I and 834	goat I and 834
I and I 835	and I will 836	I will eat 837	will eat them 838	eat them in 839	them in the 840	in the rain 841
the rain I 842	rain I and 843	I and in 844	and in the 845	in the dark 846	the dark I 847	dark I and 848

and on 849	and on a 850	on a train 851	a train train 852	and in 853	and in a 854	in a car 855
a car 857	car and 858	and in 859	and in a 860	in a tree 861	a tree tree 862	they are 864
they are 865	are so good 866	so good 867	good so 868	so good 869	so good 870	good you 871
you see 872	you see so 873	see so I 874	so I will 875	I will eat 876	will eat them 877	eat them in 878
them in a 879	in a box 880	a box box 881	box and 882	and I 883	and I will 884	I will eat 885
will eat them 886	eat them with 887	them with a 888	with a fox 889	a fox fox 890	fox and 891	and I 892
and I will 893	I will eat 894	will eat them 895	eat them in 896	them in a 897	in a house 898	
a house 899	house and 900	and I 901	and I will 902	I will eat 903	will eat them 904	eat them with 905
them with a 906	with a mouse 907	a mouse mouse 908	mouse and 909	and I 910	and I will 911	
I will eat 912	will eat them 913	eat them here 914	them here and 915	here and there 916	and there 917	
there say 918	say I 919	say I will 920	I will eat 921	will eat them 922	eat them anywhere 923	
them anywhere I 924	anywhere I do 925	I do so 926	do so like 927	so like green 928	like green eggs 929	

green eggs and 930	eggs and ham 931	and ham Thank 932	ham Thank you 933	Thank you Thank 934	
you Thank you 935	Thank you 👤 936	you 🗨️ Sam 937	🗨️ Sam I 938	Sam I am 939	I am 🗨️ 940